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Use of the normalcy index for the assessment of abnormal gait in the anterior cruciate ligament deficiency combined with meniscus injury

Xiaode Liu^a, Hongshi Huang^b, Shuang Ren^b, Qiguo Rong^a and Yingfang Ao^b

^aDepartment of Mechanics and Engineering Science, College of Engineering, Peking University, Beijing, China; ^bBeijing Key Laboratory of Sports Injuries, Institute of Sports Medicine, Peking University Third Hospital, Beijing, China

ABSTRACT

The normalcy index (NI) has been implemented by several studies as a simple index for quantitatively analyzing different gait abnormalities, such as children with cerebral palsy and idiopathic toe-walkers. However, whether the NI can be used in anterior cruciate ligament (ACL) deficiency with different types of meniscus injuries or not, has not been reported yet. In this study, 25 patients who combined different types of ACL and meniscus injuries were evaluated by the NI analysis. 12 healthy subjects were used to define the normal range of NI. The result showed that NI values of patients were significantly larger than the control group ($P < 0.05$). Meanwhile, the tendency of increasing NI values associated with increasing pathology were significant with only 5 subjects in the smallest group (Jonkheere-Terpstra test: $P < 0.001$). These results indicated that the NI was a concise yet effective tool to evaluate combined ACL and meniscus injury patients. Increasing severity degree of meniscus tears in ACL rupture patients is corresponded to increasing NI values. It also demonstrates that the proposed NI can be applied as a robustness factor to detect the discrepancy between healthy and patient subjects clinically, and has the potential in the quantitative evaluation of pre- or post-surgery and rehabilitation.

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Gait analysis; normalcy index; anterior cruciate ligament deficiency; meniscus injury

Introduction

Gait analysis is widely used as an effective tool for the evaluation of abnormal gait for the patient who have combined anterior cruciate ligament (ACL) and meniscus tears during level walking (Berchuck et al. 1990; Hurd and Snyder-Mackler 2007; Fuentes et al. 2011; Ren et al. 2018). Some typical gait patterns of patients affected by ACL and meniscus injuries, such as the stiffening strategy (Hurd and Snyder-Mackler 2007), quadriceps avoidance (Berchuck et al. 1990) and pivot-shift avoidance (Fuentes et al. 2011), have been well studied. Based on the 3-D motion capture system, traditional gait analysis provides an insight into joint kinematic and kinetic alterations, by which specific features of the abnormal gait could be quantified. However, gait analyses often yield large volume of data, and the correlated parameters make gait deviations difficult to be objectively interpreted. For example, peak knee flexion is dependent on walking speed, knee flexion moment depends on the knee flexion angle, and knee rotations in coronal and transverse planes are often coupled. Although direct comparisons of gait data have been well documented to indicate the alterations in specific

gait variables between healthy and patient subjects, a high degree of coupled parameters in lower extremity is often ignored.

An effective method to overcome this deficiency is the data reduction based on multivariate statistical methods. For instance, principal component analysis (PCA) is widely used for dimensionality reduction and feature extraction (Deluzio and Astephen 2007). Based on multiple parameters extracted from gait data, many concise indices were put forward to identify the walking patterns of both normal (Tingley et al. 2002) and abnormal gaits (Schutte et al. 2000; Romei et al. 2004; Shin et al. 2010; Leporace et al. 2012). These studies indicated that indices based on the PCA method can be an objective evaluation not only for a more general representation for both normal and pathology gait but also for the clinical relevance to assist diagnosis. In particular, Schutte (Schutte et al. 2000) proposed a normalcy index (NI) to reflect gait deviations from the average of normal gait. It was firstly used to evaluate the degree of pathological gait affected by cerebral palsy, and its utility was verified by several studies (Romei et al.

Table 1. Subject characteristics in each group.

Parameters	Control	ACLD	ACLDL	ACLDM	ACLDML
Number	12	8	5	5	7
Age (years)	29.58 ± 5.39	27.83 ± 2.23	26.60 ± 3.85	29.60 ± 5.41	27.71 ± 6.75
BMI (kg/m ²)	25.28 ± 3.03	24.25 ± 2.00	27.26 ± 5.82	26.97 ± 3.01	25.64 ± 3.29
Pace(m/s)	1.25 ± 0.10	1.23 ± 0.09	1.27 ± 0.13	1.26 ± 0.09	1.31 ± 0.16
Time since injury (months)	/	9.83 ± 7.39	9.60 ± 3.29	15.40 ± 8.11	14.14 ± 5.37

2004; Assi et al. 2006; Shin et al. 2010). Among these, no researches have involved in the assessment of ACL rupture combined with meniscus injury. Besides, although spatiotemporal and kinematic parameters were contained in the variable selection, kinetic performances were not well-considered.

Previous studies have reported that ACLD with (Zhang et al. 2016; Ren et al. 2018) or without meniscus injuries (Berchuck et al. 1990; Hurd and Snyder-Mackler 2007; Fuentes et al. 2011) could influence kinematics and kinetics of knee joint and its stability in lower extremities. The findings provide important guidelines for clinical diagnosis as well as specific treatment protocols on gait of ACL-reconstruction patients. Recently, Mok et al. (2019) performed a retrospectively analysis on ACL-reconstruction surgery and found three indices, including longer time to surgery, male sex and higher BMI, were related to a higher prevalence of medial meniscus tears. However, there is lack of a clinical meaningful indicator to accurately represent the degree of gait abnormality in ACLD with meniscus injury patients. Even with the help of gait analysis approaches, objectively quantifying gait changes from normal gait has not been well studied by now.

The aim of this study was: (1) to implement the method to calculate the normalcy index (NI) based on the selected variables and (2) to assess the method effectiveness in clinical cases by evaluating the difference of NI between control and different types of ACL and meniscus injury groups. We hypothesized that: (1) there is a significant difference in the NI magnitude between control and patient groups; (2) the NI magnitude vary from different subjects grouped by different types of meniscus injury patterns.

Method

Patients with unilateral ACL rupture and different types of meniscus injuries (diagnosed by clinical examination and MRI) were recruited. In total, 25 patients were divided into 4 groups according to the injury patterns: 8 had isolated ACL injuries (ACLD group), 5 had ACL and lateral meniscus injuries (ACLDL group), 5 had ACL and medial meniscus injuries (ACLDM

group) and 7 had ACL and medial and lateral meniscus injuries (ACLDML group). 12 normal walkers participated in this gait data collection (control group). The study was approved by the university's ethics committee, and written informed consent was obtained from all participants before the data collection. All subject characteristics are shown in Table 1.

A motion capture system (Vicon MX; Oxford Metrics, Yarnton, Oxfordshire, UK) was used to collect the gait data. There were eight cameras with a sample frequency of 100 Hz to track the subject motions. Coordinate data were filtered by a low-pass butter-worth filter at 12 Hz. Ground-reaction forces were measured using two embedded force plates (Advanced Mechanical Technology Inc., Watertown, MA, USA, 1000 Hz). Based on the validated plug-in-gait model, light-reflecting markers were used to attach to their surface skin of anatomical locations, including the medial and lateral femoral epicondyles, anterior and posterior superior iliac spine, medial and lateral sides of the calcaneus, malleoli, the frontal and lateral aspects of the thigh and the shank, posterior of the calcaneus, the first, second, and fifth metatarsal bones, the first metatarsal bone, navicular, and hallux, to track the trajectories when performing the experimental task. Every subject was asked to walk along a 10-m walkway at a self-selected speed. Five successful trials were collected for each participant.

Based on the experimental data, a commercial software AnyBody (version 6.0.5, AnyBody™ Technology, Aalborg, Denmark) was used to simulate the musculoskeletal dynamic process. The validated University of Twente Lower Extremity model was hired to calculate muscle forces and joint moments (Damsgaard et al. 2006). This model consists of 12 body segments: HAT (head, arms and trunk), pelvis, and left and right femur, patella, tibia, talus, and foot. 11 joints are defined to connect the segments, which are L5S1 and left and right hip, knee, femur, talocrural and subtalar joints. During the modeling simulation, the L5S1 and hip joints are modeled as a spherical joint with 3 degrees of freedom, and the knee, talocrural, and subtalar joints are defined as a hinge joint with only 1 degree of freedom. 159 Hill-type muscle models, which consist of 55 muscle-tendon units, are considered for

the mechanical effect. For each muscle model, the segments and isometric muscle strength are scaled by using a length-mass-fat law (Lund et al. 2015) based on the subject's morphological parameters (the height, body weight). A min/max recruitment principle solver in the software was adopted to solve the muscle redundancy problem (Marra et al. 2015). It minimizes the maximal muscle activity, and has advantages of better numerical convergence and physiological representation compared to other muscle recruitment optimization (Damsgaard et al. 2006). The principle can be simply formulated as follows:

$$\begin{aligned} &\text{Minimize } \max \left(\frac{f_i^{(M)}}{N_i} \right) \\ &\text{Subject to } \mathbf{C}\mathbf{f} = \mathbf{d} \\ &0 \leq f_i^{(M)} \leq N_i, \quad i \in \{1, \dots, n^{(M)}\} \end{aligned}$$

where $f_i^{(M)}$ represents the muscle forces and N_i refers to the strength of the muscle. $\mathbf{C}$ is the coefficient-matrix for the unknown forces and the right-hand side and $\mathbf{d}$ contains all known applied loads and inertia forces. $\mathbf{f} = [\mathbf{f}^{(M)} \quad \mathbf{f}^{(R)}]^T$, which represents muscle forces and joint reactions, respectively.

Five different gait trials were simulated for each subject, and average values were calculated as the initial data. Using a cubic spline interpolation, the initial data were normalized to 101 discrete points corresponding to 0–100% of gait cycle for the kinematic component and 61 discrete points corresponding to 0–60% of stance phase for the kinetic component, respectively. The moment and force peaks as well as force impulses during gait were calculated.

20 gait parameters were extracted from the subjects' normalized gait data. 3 kinematic and 17 kinetic parameters that are common used by previous studies were selected to perform this analysis. These are: maximum flexion angle throughout gait cycle (Shin et al. 2010), the first flexion angle valley during stance phase (Hurd and Snyder-Mackler 2007; Fuentes et al. 2011), the range of knee flexion (Ren et al. 2018), three knee contact force (mediolateral, anteroposterior and proximodistal force) peaks (Bowersock et al. 2017) during gait cycle and their corresponding phases, three knee moment (internal rotation, adduction and flexion moment) peaks during gait cycle (Deluzio and Astephen 2007; Hurwitz et al. 2010) and their corresponding phases (Deluzio and Astephen 2007), extension moment peak during terminal stance (Thambyah et al. 2004; Hurd and Snyder-Mackler 2007) and its corresponding phase and three force (mediolateral, anteroposterior and proximodistal) impulses during

the stance (Bowersock et al. 2017). The parameters were used to calculate the NI for each subject.

The NI method can be simplified as the square of the distance between the patient's parameters and the average value of normal subjects redefined by PCA method in a new orthogonal space (Schutte et al. 2000). The calculation process was outlined as follows:

Firstly, define the reference parameters of normal gait (the data from control group):

- (1) Let x_i , $i = 1, N$ represent N discrete variables from the gait cycle ($N=20$ in this study).
- (2) Let μ_i , σ_i represent the mean and standard deviation of x_i from normal subjects.
- (3) Standardize the gait variables by z_i ,

$$z_i = (x_i - \mu_i) / \sigma_i \quad i = 1, N$$

Note that z_i has the same dimension as x_i , and all data have the same unit.

Using the standardized discrete variables z_i to calculate the covariance matrix (C_{ij}).

Calculate the N eigenvalue-eigenvector pairs for the covariance matrix (C_{ij}).

Calculate a new set of scaled variables y_i , defined by

$$y_i = \left(\frac{1}{\lambda_i} \right) \sum_{j=1}^N \alpha_j z_j$$

where λ_i is the square root of the corresponding eigenvalues, α_j is the loading values determined by the principal component analysis. Then, for the $\tilde{x}_i$ discrete variables measured on patient group, the standardization was performed by using the μ_i , σ_i from the step (2),

$$\tilde{z}_i = (\tilde{x}_i - \mu_i) / \sigma_i \quad i = 1, N$$

Calculate a new set of scaled variables $\tilde{y}_i$, defined by

$$\tilde{y}_i = \left(\frac{1}{\lambda_i} \right) \sum_{j=1}^N \alpha_j \tilde{z}_j$$

Find the square of the Euclidian length for a given subject

$$d = \sum_{j=1}^N \tilde{y}_j^2$$

The number d is called 'normalcy index'.

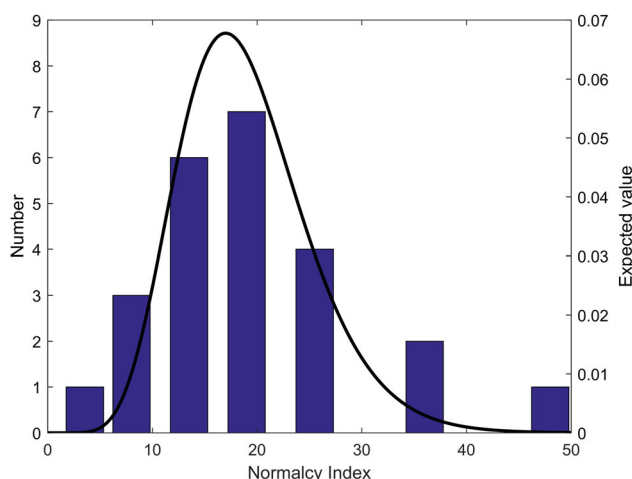


Figure 1. The theoretical and calculated distributions of NI values based on the normal subjects. Solid line represented expected χ^2 distribution with 19 degrees of freedom. The histogram showed the number of different range of NI distribution for the left and right legs.

As for the control group, both right and left sides were used to calculate the NI. Mean values and ranges for each of groups were calculated based on the NI calculation procedure. Independent sample *t*-tests were used to perform the post-hoc analysis between each of two groups. The significant difference was set at $P < 0.05$. The Jonkheere-Terpsta test was used to determine whether or not the NI ranked the groups of subjects by increasing the severity degrees of injury. There is an initial hypothesis that the mean NI values were ranked according to the severity of injury: control, ACLD, ACLDL, ACLDM and ACLDML group.

Result

All subjects' anthropometrics as well as the walking speed are showed in Table 1. Student's *t*-tests were performed and no statistical difference was found between each of two groups.

The distribution of NI for the control group was compared to the χ^2 distribution. For a 20 variables data set, it is theoretically described by a distribution curve of 19 degrees of freedom (Figure 1) (Box 1978). This distribution showed that 99.5% of the predictive values were below 39, and the value at 17 was corresponding to maximum expected value. In this study, the normal range of NI was assembled by 24 legs of 12 control subjects. The result showed that 23 of 24 NI values for the control group was within the range of 5 and 39. Meanwhile, 7 NI values corresponding to 7 normal subjects were found from 15.38 to 20.60 (the highest in the histogram in Figure 1), which was closely in agreement with the theoretical predictions.

Table 2. The NI values for each group.

Group	Number	Mean normalcy index (range)
Control	12	18.33 (3.54–48.01) ^a
ACLD	8	32.83 (15.86–66.96)
ACLDL	5	38.69 (21.17–59.70)
ACLDM	5	61.07 (42.05–88.06)
ACLDML	7	68.45 (30.19–149.23)

^aThe value of mean $\pm$ standard deviation was 18.33 ± 9.57 for the left leg indices, and it was 18.33 ± 12.45 for the right leg indices.

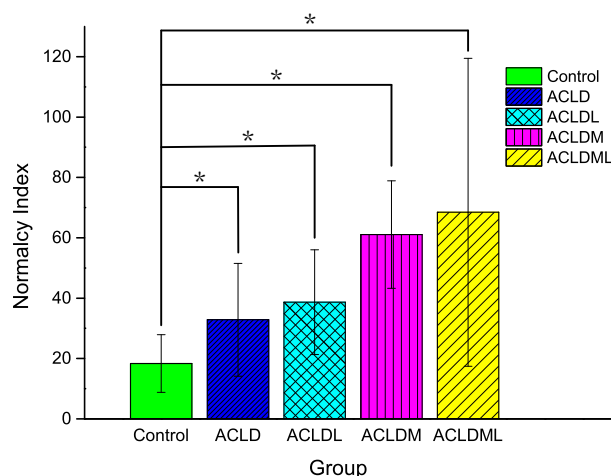


Figure 2. Comparisons for each of two groups. The group means and standard deviations are shown (*Statistically significant difference at $P < 0.05$).

The mean NI value for the control group was the lowest compared to the other four groups, and the range of NI values for the ACLDML group was largest (Table 2). There was a significant difference between the control and other patient groups (Figure 2, $P < 0.05$). Meanwhile, it was also found that the ranking of NI values from lowest to highest was: control < ACLD < ACLDL < ACLDM < ACLDML. The Jonkheere-Terpsta test was then performed and showed that the rank according to the NI values were significant ($P < 0.001$). Patients with ACLD tended to exhibited higher NI values. However, no interpatient significant difference of NI values was found in the ACLD, ACLDL, ACLDM and ACLDML groups, although $P = 0.079$ was detected between ACLDL and ACLDML groups.

Discussion

Based on 20 kinematic and kinetic parameters extracted from the gait cycle, this study quantitatively evaluated ACLD patients with or without meniscus injuries by using the NI method. The result of this study supported the first hypothesis that NI values between control and the other patient groups were significant. Although significant abnormal features have been reported by previous studies both in ACLD (Berchuck et al. 1990; Hurd and Snyder-Mackler

2007) and combined with meniscus injury patients (Zhang et al. 2016; Ren et al. 2018), few researches have considered an indexing technique to classify pathological effects on gait outcomes. As far as we know, this is the first paper that focuses on the quantitative evaluation of ACLD with or without meniscus injury patients by using the normalcy index. The most important finding was that the NI was able to quantify the gait deviations by condensing the kinematic and kinetic parameters into a simple number. Higher NI scores were corresponded to higher degrees of gait abnormality in ACL combined with meniscus injuries patients.

The expected χ^2 distribution and actual distribution extracted from the control group were similar with the previous studies who put forward the NI method (Schutte et al. 2000) and verified in the cerebral palsy patients (Romei et al. 2004; Shin et al. 2010), which confirmed the correctness of NI procedure in this study. Considering that the calculation procedure was exactly the same, and part of parameters, for instance, knee flexion kinematics, were all selected as the variables, direct comparison was conducted between this study for ACLD combined with different types of meniscus injury patients and the other two studies for cerebral palsy in children. It is notable that NI scores were lower when compared with Romei's results (15.86–149.23 for the ACLD-affected combined with meniscus injury vs 44.7–1827 for children affected by cerebral palsy), though statistical significance was both detected in these two studies. Different clinic pathological characteristics can provide a support for the results. As for the cerebral palsy patients, the biomechanical characteristics of lower limbs is significantly different from normal people, even in the level walking. It also reported that some of the participants required walking aids when performing the gait test (Romei et al. 2004). Cerebral palsy patients exhibited higher degrees of severity of abnormal gait compared with the chronic ACLD patients. In another research (Shin et al. 2010), the range of NI value was 26.75–199.74, similar to our results. The high similarity could be explained by two reasons: first, none of the patients in their work were dependent ambulators, which implied that the participant didn't require any assistance when undergo gait tests. They were more easier to finish the walking task compared with some patients in Romei's study; second, there were some similar gait variables, resulting in quite close scores based on the same method.

Interpatient NI differences in ACLD patients, either with or without meniscus tears, have also been

considered. The result of Jonkheere-Terpsta test showed that higher NI values corresponded to higher severity degrees of injury. For example, the mean score of ACLDM group was higher than that of ACLDL, which implied that there were more abnormal features for ACLDM patients compared with ACLDL patients. The function of lateral meniscus and medial meniscus on the knee joint mobility may provide an explanation of the inference. The lateral meniscus was reportedly more susceptible to the unusual compression and shear forces when performing pivot shift movement (Smith and Barrett 2001), while the medial meniscus is tightly restraint to anteroposterior tibial translation, becoming a wedge between the femur and tibia to stabilize the knee function (Levy et al. 1982). Lateral meniscus is more sensitive to the transverse and frontal movement, playing an important role in postural stability (Lee et al. 2018), while medial meniscus is more susceptible to the sagittal motion. On the other hand, the meniscus transmits 50% to 85% compressive loading through the posterior horns during flexion and extension (Fox et al. 2012). It is well distributed when the meniscus is normal. Total lateral meniscectomy of lateral meniscus results in a 40% to 50% reduction in femoral condyle contact area, while removal of medial meniscus results in a 50% to 70% decrease in contact area (Fox et al. 2012). More increasing in the load per unit area of medial meniscus may contribute to more articular cartilage damage and degeneration than that of lateral meniscus.

However, considering the wide range of NI values in each group, no clear conclusion could be made in terms of the relationship between each diagnosis category in ACLD combined with meniscus injury patients and NI scores. In our opinion, the NI value is likely less important to evaluate the exact meniscus injury patterns, as the overlap between the patient groups showed that a given NI may come from one of the four different groups. However, this index can be used to estimate the severity degrees of combined ACL and meniscus injury patients, as the trend that increasing severity degree of meniscus tears in ACL rupture patients corresponded to increasing NI values was significant within the five groups.

Traditional clinical diagnosis evaluates the degree of gait abnormality in ACLD combined with meniscus injury patients mainly by comparing alterations of gait pattern and function with the normal people, and the criteria for treatments are generally whether the knee function has restored to normal. The method in this study provides a novel and objective index by

condensing the discrete kinematic and kinetic parameters of the knee joint into a simple number. It is not necessary to indicate the status of knee function or the stage of rehabilitation after ACL-reconstruction. Just choosing a set of representative gait variables as input for the calculation makes this method more reliable and applicable in clinical evaluations.

Additionally, previous studies (Schutte et al. 2000; Romei et al. 2004; Shin et al. 2010) used only kinematic variables in order to contain the subject who needed the walking aids, leaving a limitation of excluding kinetic variables. In this study, both kinematic and kinetic variables were considered for the NI in ACLD combined with meniscus injury patients. The result showed that the NI was robust enough to detect the different pathology by the group means, despite of the overlapped ranges of NI. However, it was noteworthy that these gait parameters were extracted from the musculoskeletal modeling based on experimental data. Errors of the modelling process as well as gait measurement (e.g. the simplification of musculoskeletal model system, skin motion artificial errors) were inevitable. Optimized data set of variables, choosing hip and ankle joint parameters and a combination of other multivariate statistical methods, could further improve the adaptability and applicability of this tool.

There are some limitations in this study. First, sagittal motion were considered for the main motion during the simulation process, as the knee joint was regarded as a hinge joint. Second, the post-injury activity level (i.e. coper or non-coper status) could mask alterations in the knee joint kinematics and kinetics, which may result in different NI scores. Third, the sample size for the patient groups was relatively small. More gait data embedded in the calculation of NI as well as the division of groups by the post-injury period would be necessary to support our conclusion in the next work. Despite varies influence from gait data, the NI index exhibits high robustness to detect pathology, and provides an objective method to evaluate the pathological abnormalities.

Conclusion

This study applied the PCA-based NI method to quantitatively evaluate the differences between ACLD with or without meniscus injury patients and control people. The distributions of NI scores showed that: (1) there were statistical significant differences between ACLD-affected patients with different types of meniscus injuries and control subjects. The NI was

capable of quantifying gait alterations by the scores, which had clinical relevance to assist in diagnosis of abnormal gait and rehabilitation after ACL-reconstruction; (2) the trend of higher NI scores corresponding to higher potential pathology in patients who had combined ACL and meniscus injuries was obvious by using the Jonkheere-Terpsta test. It implied that different types of pathology reflected different range of NI scores, which theoretically provided a basis for the classification of sub-groups for a specific injury, for example, different meniscal zones of medial and lateral meniscus (Smith and Barrett 2001). The normalcy index is a simple yet effective tool to evaluate movement disorders. It has the potential to be an excellent indicator to identify the abnormal gait associated with different types of injuries in lower limbs.

Disclosure statement

No potential conflict of interest was reported by the authors.

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